

DETAILED DESCRIPTION OF THE INVENTION

Expanded Sensor Array and Security Architecture

In various embodiments, the penile geometry profiling device may incorporate multiple time-of-flight (ToF) ranging sensors arranged in defined spatial configurations to enhance surface sampling density and reconstruction fidelity. The device may include between one and ten discrete ToF sensors positioned longitudinally, circumferentially, angularly offset, or in distributed array patterns within the housing structure. Multiple ToF sensors may operate sequentially or concurrently to capture radial distance measurements from different positions relative to the anatomical surface.

In certain embodiments, the device may incorporate multi-zone or array-based ToF sensing modules capable of capturing multiple depth points within a defined sensing field rather than a single-point distance measurement. Such multi-zone ToF modules may enable localized surface contour mapping within a single sampling interval, thereby increasing spatial resolution and reducing dependence on longitudinal displacement for reconstruction completeness.

The optical motion sensing subsystem may likewise comprise one or more optical flow sensors positioned along the measurement pathway. Multiple motion sensors may be configured to detect displacement across different contact or near-contact regions to improve positional accuracy and reduce motion drift. In some embodiments, optical motion data may be fused with inertial measurement data to generate a corrected longitudinal displacement model.

To enhance measurement authenticity and reduce spoofing risk, the device may further incorporate one or more physiological verification sensors. Such sensors may include a pressure sensor configured to detect tissue compliance characteristics, a photoplethysmography (PPG) sensor configured to detect vascular pulsatility, or other bio-signal detection components capable of confirming the presence of living tissue during scanning. Anti-spoofing logic may analyze dynamic physiological signals in conjunction with geometric sampling data to validate measurement integrity.

In certain embodiments, the device may further include onboard encryption hardware configured to secure acquired measurement data prior to transmission or storage. Encryption may be implemented at the firmware or hardware level using symmetric or asymmetric cryptographic protocols. Data may be encrypted during acquisition, during transmission to external devices, or during storage in local memory. In some implementations, unique device-level cryptographic keys may be provisioned to protect user-specific anatomical geometry profiles.

The multi-sensor, anti-spoofing, and encryption architectures described herein may be implemented individually or in combination and are not limited to any specific sensor count, arrangement, cryptographic protocol, or verification methodology, provided that the system remains capable of acquiring radial distance measurements and correlated positional data sufficient to generate a user specific penile anatomical geometry profile.